

Belle River

Shoreline length: 62 km



Shore Types

Belle River is an exposed stretch of diverse coast characterized mainly by wetlands and low plains along the south-east end of PEI.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
24	8	2	10	40	0	10	6



Dunes, sandspit and estuarine wetlands north of Big Point

Governance, Infrastructure, and Economy

This SMU is almost completely located within the Rural Municipality of Belfast (~94%).

Key infrastructure includes a portion of the Trans-Canada Highway and provincial roads: Ponds Road to Ponds Road Beach (South Pinette); Gascoigne Road (South Pinette); the TCH at Bears Way, Flatt River, and Flat River to Big Point Road; Camp Road (Route 261, upper Flat River); Big Point Road at the TCH to its end; the TCH from Flat River to Belle River; Culloden Road (Route 203, Flat River at the TCH); Nicholson Road (Belle River, to the beach); Stewart Point Road (Belle River, at the coast); and Factory Road (Belle River). Coastal economic drivers are limited. Local and legacy land use constraints include unpaved coastal roads running parallel to the coast.

Natural Assets and Heritage

There are no Provincial or Federal Parks nor Wildlife Management Areas in Belle River, however Ponds Road Beach is located in this SMU.

The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

The area has relatively low built up % though some infrastructure exists close to the shore. Due to the primary shore types and exposure to waves the area experiences moderate flood and erosion hazards. The low plains, sandspits, and wetlands are vulnerable to increasing flood hazards. Erosion hazard is highest along the exposed dune.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of

Proposed SMP Strategies

No Active Intervention is proposed for the short-term, followed by Managed Realignment for the limited number of assets at risk.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including dunes, beaches, wetlands and cliff habitat.		

sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Belle River				
Flood risk	2 [<1/km]	3 [<1/km]	4 [<1/km]	12 [<1/km]
Erosion risk	NA	16 [<1/km]	20 [<1/km]	39 [1/km]

Sediment transport - The estimated net sediment transport rate averaged across the SMU is high and feeds numerous beaches and sandspits. Sediment transport rate potential is highest along the center of the SMU.



LEGEND - Shoreline Management Unit - Net Longshore Sediment Transport - High Hazard Area - Flood Extent - Present Day (100-year Return) - Flood Extent - 1.2m SLR (100-year Return) - Property Boundary - Building > 90ft²	 0 250 500 1,000 Metres Coordinate System: NAD 83 CSRS PEI Double Stereographic NOTES:		COASTAL EROSION AND FLOOD HAZARD Shoreline Management Unit: Belle River	<i>Prince Edward Island</i> Île-du-Prince-Édouard CBCL
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